

Introduction to Database

1. Which schema describes the database design at the physical level?

- A) Logical Schema
- B) Physical Schema
- C) View Schema
- D) External Schema

Answer: B) Physical Schema

2. In the relational model, the number of attributes in a table is termed as:

- A) Cardinality
- B) Degree
- C) Domain
- D) Tuple

Answer: B) Degree

3. Which normal form is based on the concept of full functional dependency?

- A) First Normal Form (1NF)
- B) Second Normal Form (2NF)
- C) Third Normal Form (3NF)
- D) Fourth Normal Form (4NF)

Answer: B) Second Normal Form (2NF)

4. Which of the following is NOT a property of a transaction (ACID)?

- A) Atomicity
- B) Consistency
- C) Integrity
- D) Durability

Answer: C) Integrity (The I stands for Isolation)

5. A logical unit of work that must be either wholly committed or wholly rolled back is called:

A) A Transaction

B) A State

C) A Query

D) A Session

Answer: A) A Transaction

6. What is the main purpose of normalization?

A) To increase data redundancy

B) To minimize data redundancy and insertion anomalies

C) To increase the number of tables indiscriminately

D) To make querying slower

Answer: B) To minimize data redundancy and insertion anomalies

7. Which SQL command is used to retrieve data from a database?

A) GET

B) OPEN

C) SELECT

D) FETCH

Answer: C) SELECT

8. The primary key must contain:

A) Unique and Not Null values

B) Unique and Null values

C) Duplicate and Not Null values

D) Duplicate and Null values

Answer: A) Unique and Not Null values

9. Which key is used to establish a relationship between two tables?

A) Candidate Key

B) Foreign Key

C) Super Key

D) Alternate Key

Answer: B) Foreign Key

10. A set of attributes that can uniquely identify a tuple in a relation is called a:

A) Super Key

B) Domain

C) Schema

D) Instance

Answer: A) Super Key

11. Which diagrammatic technique is used for database design?

A) Flowchart

B) E-R Model

C) Data Flow Diagram

D) Use Case Diagram

Answer: B) E-R Model

12. In an E-R diagram, a double rectangle represents:

A) Strong Entity

B) Weak Entity

C) Relationship

D) Composite Attribute

Answer: B) Weak Entity

13. The constraint "A value in the foreign key must exist in the referenced primary key" is known as:

A) Entity Integrity

B) Domain Integrity

C) Referential Integrity

D) User-defined Integrity

Answer: C) Referential Integrity

14. Which command is used to remove all records from a table, but leaves the table structure intact?

- A) DELETE
- B) DROP
- C) TRUNCATE
- D) ERASE

Answer: C) TRUNCATE

15. What does the `DISTINCT` keyword do in an SQL query?

- A) Sorts the result
- B) Limits the result rows
- C) Eliminates duplicate rows from the result
- D) Joins two tables

Answer: C) Eliminates duplicate rows from the result

16. A transaction completes its execution is said to be:

- A) Committed
- B) Aborted
- C) Rolled back
- D) Failed

Answer: A) Committed

17. Which of the following is an aggregate function in SQL?

- A) SELECT
- B) COUNT
- C) WHERE
- D) ORDER BY

Answer: B) COUNT

18. The specific representation of data as it is stored on the storage media is the concern of the:

- A) External level
- B) Conceptual level

C) Internal (Physical) level

D) View level

Answer: C) Internal (Physical) level

19. An entity that depends on another entity for its existence is called a:

A) Strong Entity

B) Weak Entity

C) Variant Entity

D) Dependent Entity

Answer: B) Weak Entity

20. Which of the following is NOT a valid SQL data type?

A) CHAR

B) VARCHAR

C) DECIMAL

D) CURRENCY

Answer: D) CURRENCY (Commonly MONEY or DECIMAL depending on DB, Currency is an Access type but not standard SQL)

21. The command used to modify the structure of an existing table is:

A) UPDATE

B) ALTER

C) MODIFY

D) CHANGE

Answer: B) ALTER

22. The condition that a database is in a state where all integrity constraints are satisfied is called:

A) Isolation

B) Consistency

C) Atomicity

D) Durability

Answer: B) Consistency

23. Which join returns all rows from the left table and the matching rows from the right table?

- A) Inner Join
- B) Right Outer Join
- C) Left Outer Join
- D) Full Outer Join

Answer: C) Left Outer Join

24. A deadlock exists in a system when:

- A) Two transactions are waiting for each other to release a lock
- B) A transaction fails
- C) The system crashes
- D) A transaction takes too long

Answer: A) Two transactions are waiting for each other to release a lock

25. The process of managing simultaneous operations on the database without them interfering with one another is called:

- A) Serializability
- B) Recovery
- C) Concurrency Control
- D) Backup

Answer: C) Concurrency Control

26. Which of the following is a TCL (Transaction Control Language) command?

- A) GRANT
- B) REVOKE
- C) ROLLBACK
- D) CREATE

Answer: C) ROLLBACK

27. In relational algebra, the symbol σ (Sigma) represents:

- A) Projection
- B) Selection
- C) Cartesian Product
- D) Division

Answer: B) Selection

28. In relational algebra, the symbol π (Pi) represents:

- A) Selection
- B) Projection
- C) Join
- D) Union

Answer: B) Projection

29. Which normal form deals with Transitive Dependency?

- A) 1NF
- B) 2NF
- C) 3NF
- D) 4NF

Answer: C) 3NF

30. A table is in BCNF if it is in 3NF and:

- A) Every non-prime attribute is dependent on the primary key
- B) For every functional dependency $X \rightarrow Y$, X is a super key
- C) It has no composite keys
- D) It has no multi-valued dependencies

Answer: B) For every functional dependency $X \rightarrow Y$, X is a super key

31. Data about the database structure (schema) is stored in the:

- A) Data File
- B) System Catalog / Data Dictionary

C) Log File

D) Buffer

Answer: B) System Catalog / Data Dictionary

32. Which clause is used to filter groups in an aggregate query?

A) WHERE

B) HAVING

C) GROUP BY

D) ORDER BY

Answer: B) HAVING

33. Which of the following allows a user to undo changes that have not yet been committed?

A) COMMIT

B) SAVEPOINT

C) ROLLBACK

D) GRANT

Answer: C) ROLLBACK

34. A functional dependency $A \rightarrow B$ is trivial if:

A) B is a subset of A

B) A is a subset of B

C) A and B are disjoint

D) A is a primary key

Answer: A) B is a subset of A

35. Checkpoints are used in a database for:

A) Query Optimization

B) Recovery

C) Security

D) Concurrency

Answer: B) Recovery

36. Which file organization provides the fastest random access to records?

- A) Sequential
- B) Heap
- C) Hashing
- D) Clustered

Answer: C) Hashing

37. A database trigger is:

- A) A procedure that runs automatically when a specific event occurs
- B) A button in the GUI
- C) A type of index
- D) A backup mechanism

Answer: A) A procedure that runs automatically when a specific event occurs

38. Which operator is used to search for a specified pattern in a column?

- A) LIKE
- B) IN
- C) BETWEEN
- D) EXISTS

Answer: A) LIKE

39. The 'wildcard' character representing zero, one, or multiple characters in SQL is:

- A) ?
- B) _ (underscore)
- C) % (percent)
- D) *

Answer: C) % (percent)

40. What is the result of a Cartesian Product between a table with 5 rows and a table with 4 rows?

- A) 9 rows

B) 20 rows

C) 5 rows

D) 4 rows

Answer: B) 20 rows

41. Which command is used to add new rows to a table?

A) ADD

B) INSERT

C) UPDATE

D) APPEND

Answer: B) INSERT

42. What is an 'Instance' of a database?

A) The overall design of the database

B) The state of the database at a specific point in time

C) The capacity of the database

D) The hardware running the database

Answer: B) The state of the database at a specific point in time

43. Which type of user writes specialized database applications that do not fit into the traditional data processing framework?

A) Naive Users

B) Sophisticated Users

C) Specialized Users

D) Stand-alone Users

Answer: C) Specialized Users

44. Which level of abstraction describes only part of the entire database to the user?

A) Conceptual Level

B) Physical Level

C) View Level (External Level)

D) Storage Level

Answer: C) View Level (External Level)

45. Relationships are represented in the Relational Model by:

A) Pointers

B) Links

C) Foreign Keys

D) Objects

Answer: C) Foreign Keys

46. The command `DROP TABLE` does what?

A) Deletes the table definition and all its data

B) Deletes only the data

C) Deletes the table but saves the structure

D) Archives the table

Answer: A) Deletes the table definition and all its data

47. Granting `SELECT` permission to a user is an example of:

A) Integrity Constraint

B) Authorization / Security

C) Recovery

D) Concurrency

Answer: B) Authorization / Security

48. A unique constraint ensures that:

A) No two rows have the same value in the specified column(s)

B) The column cannot be NULL

C) The column is a Foreign Key

D) The value must be an integer

Answer: A) No two rows have the same value in the specified column(s)

49. Which of the following is NOT a database object?

- A) Table
- B) View
- C) Index
- D) Tuples

Answer: D) Tuples (Tuple is a row of data, not a schema object like Table or Index)

50. The "I" in ACID stands for:

- A) Integrity
- B) Isolation
- C) Immediate
- D) Indexing

Answer: B) Isolation

51. A schedule where the operations of transactions are executed one after another without interleaving is called a:

- A) Serial Schedule
- B) Concurrent Schedule
- C) Parallel Schedule
- D) Non-serial Schedule

Answer: A) Serial Schedule

52. Two-Phase Locking (2PL) is a protocol used for:

- A) Recovery
- B) Deadlock Prevention
- C) Concurrency Control
- D) Authentication

Answer: C) Concurrency Control

53. An index is used to:

- A) Store data

- B) Speed up data retrieval
- C) Ensure data integrity
- D) Perform calculations

Answer: B) Speed up data retrieval

54. B-Tree and B+-Tree are types of:

- A) Data Models
- B) Indices
- C) Tables
- D) Queries

Answer: B) Indices

55. Which command is used to save the changes made by a transaction permanently?

- A) ROLLBACK
- B) SAVE
- C) COMMIT
- D) DONE

Answer: C) COMMIT

56. Which SQL clause is used to filter records?

- A) GROUP BY
- B) ORDER BY
- C) WHERE
- D) SELECT

Answer: C) WHERE

57. To sort the result set in descending order, which keyword is used?

- A) ASC
- B) DESC
- C) SORT
- D) DOWN

Answer: B) DESC

58. `COUNT(\*)` returns:

- A) The number of distinct values in a column
- B) The number of rows in the table including NULLs
- C) The number of columns in the table
- D) The sum of values

Answer: B) The number of rows in the table including NULLs

59. The intersection of a row and a column is called a:

- A) Data
- B) Field
- C) Value
- D) Cell

Answer: D) Cell (or Value/Field depending on context) - *Field implies the column concept, Cell implies the specific unit.* Correct term is Data Value or Field Value. Let's select "Cell" or "Field Value". Since Cell is here: D) Cell.

60. Which key allows NULL values?

- A) Primary Key
- B) Foreign Key (Unless specified Not Null)
- C) Candidate Key
- D) Super Key

Answer: B) Foreign Key

61. Which type of failure occurs when the system stops working but the disk storage remains intact?

- A) Transaction Failure
- B) System Crash
- C) Disk Failure
- D) Media Failure

Answer: B) System Crash

62. The capability of a system to recover data after a failure is called:

- A) Reliability
- B) Recoverability
- C) Availability
- D) Atomicity

Answer: B) Recoverability

63. Which normal form deals with Multi-valued Dependency?

- A) 3NF
- B) BCNF
- C) 4NF
- D) 5NF

Answer: C) 4NF

64. What is 'Data Mining'?

- A) Entering data into a database
- B) Extracting hidden patterns from large datasets
- C) Backing up data
- D) Compressing data

Answer: B) Extracting hidden patterns from large datasets

65. A 'Data Warehouse' is typically:

- A) Optimized for transaction processing (OLTP)
- B) Optimized for analytical processing (OLAP)
- C) Used for storing logs only
- D) A temporary storage

Answer: B) Optimized for analytical processing (OLAP)

66. Which operation is used to combine two or more tables based on a related column between them?

- A) JOIN

B) UNION

C) SELECT

D) LINK

Answer: A) JOIN

67. `DELETE FROM table\_name` without a `WHERE` clause will:

A) Delete the table structure

B) Delete the first row

C) Delete all rows

D) Produce an error

Answer: C) Delete all rows

68. Which of the following is NOT an aggregate function?

A) SUM()

B) AVG()

C) MAX()

D) ROUND()

Answer: D) ROUND() (It is a scalar function)

69. In E-R modeling, a diamond shape with a double border represents:

A) Strong Relationship

B) Identifying Relationship (for weak entities)

C) Recursive Relationship

D) Ternary Relationship

Answer: B) Identifying Relationship

70. The number of entities participating in a relationship is called the:

A) Degree of the relationship

B) Cardinality

C) Participation

D) Role

Answer: A) Degree of the relationship

71. A query within another query is called:

- A) Nested Query / Subquery
- B) Parallel Query
- C) Join Query
- D) Recursive Query

Answer: A) Nested Query / Subquery

72. What does the `OR` operator do?

- A) Displays a record if ALL conditions are true
- B) Displays a record if ANY of the conditions are true
- C) Displays a record if NONE of the conditions are true
- D) Combines two tables

Answer: B) Displays a record if ANY of the conditions are true

73. `UPDATE` command is used to:

- A) Add a new column
- B) Add a new row
- C) Modify existing data in a table
- D) Delete a table

Answer: C) Modify existing data in a table

74. The schema that defines the logical view of the entire database is the:

- A) Internal Schema
- B) Conceptual Schema
- C) External Schema
- D) Physical Schema

Answer: B) Conceptual Schema

75. Data independence is achieved by:

- A) Data Abstraction
- B) Data Redundancy
- C) Data Integrity
- D) Data Security

Answer: A) Data Abstraction

76. The constraint that no primary key value can be NULL is called:

- A) Referential Integrity
- B) Entity Integrity
- C) Domain Integrity
- D) Key Integrity

Answer: B) Entity Integrity

77. An attribute that uniquely identifies an entity from an entity set is called a:

- A) Key attribute
- B) Composite attribute
- C) Derived attribute
- D) Multi-valued attribute

Answer: A) Key attribute

78. The overall design of the database is called:

- A) Database Instance
- B) Database Schema
- C) Database Snapshot
- D) Database Structure

Answer: B) Database Schema

79. Which SQL constraint ensures that all values in a column are different?

- A) NOT NULL
- B) UNIQUE
- C) CHECK

D) DEFAULT

Answer: B) UNIQUE

80. `AVG()` function returns:

A) The sum of values

B) The number of values

C) The average value

D) The maximum value

Answer: C) The average value

81. Which layer of the 3-schema architecture deals with physical storage details?

A) Internal Level

B) Conceptual Level

C) External Level

D) Logical Level

Answer: A) Internal Level

82. The term "Relation" comes from:

A) English Language

B) Programming

C) Mathematical Set Theory

D) Physics

Answer: C) Mathematical Set Theory

83. Which of the following is true about a "View"?

A) It physically stores data

B) It is a virtual table derived from other tables

C) It cannot be queried

D) It takes up significant storage space

Answer: B) It is a virtual table derived from other tables

84. `GRANT` is a command of:

- A) DDL
- B) DML
- C) DCL
- D) TCL

Answer: C) DCL

85. Which normal form removes partial dependencies?

- A) 1NF
- B) 2NF
- C) 3NF
- D) BCNF

Answer: B) 2NF

86. Which SQL keyword is used to retrieve only unique values?

- A) UNIQUE
- B) DISTINCT
- C) DIFFERENT
- D) SINGLE

Answer: B) DISTINCT

87. Which operator checks if a value is within a range?

- A) IN
- B) LIKE
- C) BETWEEN
- D) EXISTS

Answer: C) BETWEEN

88. The `AS` keyword in SQL is used to:

- A) Create a table
- B) Rename a column or table with an alias

C) Select all columns

D) Filter data

Answer: B) Rename a column or table with an alias

89. A transaction that has completed successfully is:

A) Partially Committed

B) Committed

C) Failed

D) Active

Answer: B) Committed

90. The collection of all the data that is stored in the database at a particular moment is known as:

A) Schema

B) Instance

C) Extension

D) Both B and C

Answer: D) Both B and C

91. If a table has two candidate keys, and one is chosen as primary key, the other is called:

A) Foreign Key

B) Alternate Key

C) Super Key

D) Compound Key

Answer: B) Alternate Key

92. The `NOT NULL` constraint ensures:

A) A column cannot have duplicate values

B) A column cannot have NULL values

C) A column must be a primary key

D) A column must be a foreign key

Answer: B) A column cannot have NULL values

93. `MIN()` function returns:

- A) The smallest value of the selected column
- B) The largest value
- C) The average value
- D) The total count

Answer: A) The smallest value of the selected column

94. To delete a database, the command is:

- A) DELETE DATABASE
- B) DROP DATABASE
- C) REMOVE DATABASE
- D) ERASE DATABASE

Answer: B) DROP DATABASE

95. In an E-R diagram, a dashed oval represents:

- A) Multi-valued attribute
- B) Derived attribute
- C) Composite attribute
- D) Key attribute

Answer: B) Derived attribute

96. Which property ensures that the execution of a transaction is isolated from other transactions?

- A) Atomicity
- B) Consistency
- C) Isolation
- D) Durability

Answer: C) Isolation

97. An unnormalized relation contains:

- A) Atomic values

B) Repeating groups

C) Primary keys

D) Foreign keys

Answer: B) Repeating groups

98. Which of the following is a unary operator?

A) Union

B) Join

C) Project

D) Intersect

Answer: C) Project

99. The "cardinality ratio" describes:

A) The number of attributes in a relation

B) The number of relationship instances an entity can participate in

C) The size of the database

D) The number of tables

Answer: B) The number of relationship instances an entity can participate in

100. Write-Ahead Logging (WAL) is a technique used for:

A) Indexing

B) Atomicity and Durability (Recovery)

C) Security

D) Query Optimization

Answer: B) Atomicity and Durability (Recovery)